

# Implantable Tibial Nerve Stimulation (ITNS)

Patient Information · A small implanted device near the ankle for overactive bladder

WARWIKI

An implantable tibial nerve stimulator is a small device placed under the skin of the lower leg, just above the ankle. It gently stimulates the **tibial nerve** to calm an **overactive bladder**: sudden urges, frequent trips and urge leaks. It works like PTNS, the weekly office treatment at the ankle. Because the device is **implanted**, you get the treatment **at home**.

## How It Works

The tibial nerve runs near the inner ankle. It links up to the spinal nerves that control the bladder. Gentle pulses to this nerve help **calm** the bladder, so you have less urgency and fewer leaks. There is no medicine, so there are no side effects in the rest of the body.

## Benefits and Risks in Numbers

about  
**6–8 in 10**  
had their urge  
leaks cut by half or  
more (results differ  
by device)

about  
**1–2 in 10**  
had a problem with  
the device or  
wound, mostly  
minor

up to  
**3 years**  
the benefit lasted  
in study follow-up  
so far

Numbers are averages from published studies. Your own results may differ; ask your team.

## Is It Right for Me?

- An option for urge leaks, especially if pills did not help, caused side effects, or you prefer not to take them. It can suit people who found office PTNS helpful but hard to keep up.
- Unlike sacral nerve stimulation, there is usually **no test phase** first.
- **Tell your team** if you have poor skin, swelling, or vein or artery disease in the lower leg, past surgery near the ankle, any metal implant in that lower leg, or if you are pregnant. Some devices are not used in men whose enlarged prostate blocks the flow of urine.

## Learn the Terms

### Overactive bladder

Urgency and frequent urination from the bladder squeezing too soon.

### Tibial nerve

A nerve near the ankle that links to the bladder's nerve supply.

### Neuromodulation

Using gentle electrical pulses to calm overactive nerves.

### Implant

A small device placed under the skin during a short procedure.

### Wearable

A removable band on the lower leg that powers some devices.

### Local anesthesia

Numbing medicine so the area does not hurt while you stay awake.

### Diathermy

A deep heat treatment some therapists use. It can harm nerve implants.

**Will it hurt?** The device is usually placed with **numbing medicine**. You are awake and may feel pressure, not pain. Some people also get sedation or other anesthesia. Most people go home the same day. The treatment itself feels like mild **tingling** or a gentle toe curl.

**First few days**

Soreness and swelling.  
Rest and raise your leg  
when you can.

**About 2 weeks**

Your wound is checked.  
Stitches, if any, come out.

**Weeks 2–4**

Most devices are turned  
on once the wound has  
healed.

**Weeks to months**

The benefit builds. Follow-  
up visits check the  
settings.

**How It's Done**

- 1 Through a **small cut above the inner ankle**, the device is placed near the tibial nerve. It takes about 20–35 minutes.
- 2 The cut is closed and left to heal.
- 3 Your team sets up the device and shows you how to use it.

**Device Types**

- **Built-in battery:** nothing to wear. It runs on its own schedule. The battery lasts some years, then the device is replaced.
- **Rechargeable:** you charge it through the skin now and then.
- **Wearable-powered:** no battery inside. You wear a band on the lower leg during home sessions.

Your team will tell you which device you have and how to use it.

**Living With It**

- **Before any MRI**, show your implant card. Some scans are not allowed. Never take a wearable, charger or magnet into the MRI room.
- Do not have **diathermy** (deep heat treatment) anywhere on your body. Tell every therapist that you have this implant.
- It can be turned off or removed if needed.

**Call the office that did your procedure if you have:**

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- Redness, swelling or fluid at the ankle wound
- New or worse pain at the implant
- A sudden loss of benefit, or stimulation that feels uncomfortable

**Go to an emergency department if:**

- You cannot reach the office and are worried

**Three Things to Remember**

1. It is a small ankle implant that gives PTNS-type treatment at home.
2. Treatment usually starts once the wound heals. The benefit builds over weeks.
3. Show your implant card before any MRI. Never have deep heat treatment (diathermy).